

Alejandro Sánchez Yalí

Senior Full Stack Engineer – React, Next.js, TypeScript, Web3



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SUMMARY

Senior engineer with 8+ years building web applications, 7+ of them in React and 6+ in Next.js and TypeScript. Started on blockchain: two years at BCFort designing Ethereum and Hyperledger platforms, implementing smart contracts and building the decentralized app frontends that talked to them with Web3.js. Since then, three years on a distributed team at Monadical, then consulting and a year at Lapzo shipping products end to end, including real-time interfaces. M.Sc. in Mathematics. I care about turning complex systems into interfaces people can trust.

TECHNICAL SKILLS

- **Frontend:** React.js (7+ years), Next.js (6+ years), TypeScript (6+ years), Tailwind CSS, Shadcn UI, Redux, Zustand, Zod, React Testing Library, Three.js / React Three Fiber
- **Blockchain:** Ethereum, smart contracts, Web3.js, decentralized app frontends, NFT marketplaces, blockchain transaction analytics
- **Real-time & APIs:** WebSockets, Server-Sent Events, Firebase Realtime Database, RESTful and GraphQL APIs
- **Backend:** Node.js, NestJS, Express, Python (Django, FastAPI), PostgreSQL, Redis, MongoDB
- **Cloud & Engineering:** AWS, GCP, Docker, Terraform, GitHub Actions, CI/CD, TDD (Vitest, Jest), code review, mentoring

EXPERIENCE

Independent AI Consultant

Oct 2026 – Present

Freelance

Remote

- Lead engineer of Aluna, an AI recruitment platform for staffing agencies: two-service architecture (Next.js 15 + stateless FastAPI), durable analysis pipeline on Inngest, multi-model routing through LiteLLM, and Ley 1581/2012 compliance built into the data model

AI Specialist

Aug 2025 – Sep 2026

Lapzo

Remote, Mexico

- Made the key architectural decisions for an educational platform, designing the integration between services and the real-time communication layer
- Implemented real-time features with Next.js, NestJS and Firebase Realtime Database
- Designed the architecture of Express courses (fast course creation) and built the audit system for courses and learning paths: change logs, exportable reports and permission-aware history
- Built the content-generation pipelines now used by the platform's learners, with Celery workers running long jobs in the background

Independent Software & AI Consultant

Aug 2024 – Jul 2025

Freelance

Remote

- Designed the database architecture for a client's workflow automation platform in an NX multi-repo, and served as code reviewer for their team
- Lead engineer of CREARIA's multi-tenant AI support agent on WhatsApp, email and Instagram, in production with several client businesses including a real-estate agency: led a team of three across 5 FastAPI services and 2 Next.js dashboards on AWS ECS with Terraform, RAG over pgvector, 11 MCP tools scoped per tenant, LiteLLM fallback routing and human escalation; load-tested to 250 virtual users. Also designed the architecture of an AI coaching agent for one of CREARIA's clients
- Ran technical design sessions with clients, translating business requirements into system architectures

Full Stack Engineer

Apr 2021 – Aug 2024

Monadical

Remote, Canada/US

- Full-stack engineer on ZoHuddle, a video conferencing app for scientific conferences (Neuromatch) that matches participants by mutual interests and supports conferences of about 200 participants: React front end and Django back end, with Django Channels (WebSockets) and Twilio Video

- Full-stack engineer on Virtue Poker, a decentralized poker platform on Ethereum supporting up to 50 concurrent tables of 8 players: React front end, Django back end and smart-contract integration
- Took part in system design and architecture decisions for new features and services, weighing trade-offs across performance, scalability and maintainability
- Designed ETL pipelines and custom Django ORM extensions; managed deployments on AWS and GCP with Docker and CI/CD
- Reviewed code on a distributed team in three countries; wrote tests with Vitest and Jest

Full Stack Engineer

BCFort

Aug 2018 – Oct 2020

Medellín, Colombia

- Led development of blockchain solutions with Ethereum and Hyperledger, including NFT marketplaces
- Built decentralized app frontends and implemented smart contracts; developed platforms with React, Node.js, Python (Django) and Web3.js
- Designed system architecture for blockchain platforms: smart contract patterns, data models and service boundaries; managed infrastructure on AWS, GCP and IBM Cloud
- Mentored junior engineers, supporting their technical growth and code quality

Professor of Mathematics & Physics (part-time)

Universidad de Antioquia

Jan 2010 – Oct 2021

Medellín, Colombia

- Taught mathematics and physics; in the last year and a half also machine learning and data science with Python and the scientific stack

FEATURED PROJECTS

- **VitaStock – Surgical Supply Chain for Clinics**  – *Python, FastAPI, Next.js, TypeScript, PostgreSQL, Docker, Railway*
 - Architect and lead engineer of a platform commissioned by a private surgical clinic in Colombia – 12 modules and 5 user roles designed to replace spreadsheet-based inventory and procurement; delivered, with rollout at the clinic pending
 - Surgery scheduling with supply baskets and real-time consumption tracking, lot-level traceability with expiry alerts, and true per-patient cost through a cardex
- **Plixiq – AI Customer Support on WhatsApp**  – *Python, FastAPI, LiteLLM, PostgreSQL, Next.js, Railway*
 - Took the product from an idea to a deployed system as architect and lead engineer: discovery, architecture, backend, frontend and deploy
 - Shipped about 46,000 lines of backend and frontend plus about 590 backend tests, deployed on plixiq.com at \$60–80/month because cost scales with LLM usage rather than infrastructure

OPEN SOURCE

- **talking-avatar-with-ai**  – 464 stars, 110 forks. Open-source conversational AI avatar: OpenAI GPT, Whisper, ElevenLabs, Rhubarb lip-sync, Three.js
- **Monadical-SAS/Morpheus**  – 19 pull requests. Open-source AI image generation platform: FastAPI, React, Stable Diffusion

TALKS & WRITING

- **Scaling Learning Models: Parallelism Strategies in JAX with Flax**  – PyCon Colombia, 2024
- **Engineering case studies:** [Plixiq](#)  · [CREARIA Agent](#)  · [Aluna](#)  – architecture and trade-offs behind each system (2026)

EDUCATION

M.Sc. in Mathematics

Universidad de Antioquia

Aug 2010 - Sep 2013

Medellín, Colombia

B.Sc. in Teaching Mathematics and Physics

Universidad de Antioquia

Jan 2004 - Oct 2009

Medellín, Colombia